Marxian Competitive Dynamics Simulator

Technical Documentation, Assumptions, and User Guide

Version 2.0

Based on Marx, Capital Vol. I (Ch. 10, 15) and Vol. III (Ch. 13–15)

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Contents

1. Purpose and Overview 4
2. Theoretical Framework 5
  2.1 The Labor Theory of Value 5
  2.2 Output as a Function of All Production Inputs 5
  2.3 Absorption Costing and Fixed Overhead 5
  2.4 Price-Taking Firms and Market Allocation 6
  2.5 The Organic Composition of Capital and the Rate of Profit 6
  2.6 The Working Day and Physiological Limits 6
3. Model Structure and Formulas 8
  3.1 Variable Capital (v) 8
  3.2 Output 8
  3.3 Capital Stock and Annual Cost 8
  3.4 Production Cost Per Widget 8
  3.5 Absorbed Cost Per Widget Sold 8
  3.6 Market Allocation 9
  3.7 Financial Metrics 9
4. Assumptions and Modeling Decisions 10
  4.1 Core Assumptions 10
  4.2 Key Modeling Decisions 10
    Absorption costing over variable costing 10
    Two distinct per-unit cost metrics 10
    No inventory stockpile 11
    Output grounded in all four production inputs 11
    Rate of profit on capital stock, not revenue 11
    Monthly charts for one operating year 11
5. User Inputs: Reference Guide 12
  5.1 Market Inputs 12
    Market price: effects of changing 12
    Market size: effects of changing 12
  5.2 Firm-Level Inputs 12
    Workers 13
    Hours per day 13
    Days per week 14
    Tech multiplier 14
    Capital lifespan 14

6. Output Metrics Reference 15
6.1 Labor 15
6.2 Capital Composition 15
6.3 Production & Market 15
6.4 Per Widget 16
6.5 Annual 16
6.6 Charts 16
7. Interpretive Notes and Scenarios 17
7.1 The Core Competitive Dynamic 17
7.2 The Underutilization Penalty 17
7.3 Volume vs. Cost Competition 17
7.4 Extending the Working Day 17
7.5 The Rate of Profit Paradox 17
7.6 The Breakeven Month 18

1. Purpose and Overview

This tool models the competitive dynamics of capitalist production as described by Karl Marx in Capital, Volumes I and III. It makes the abstract mechanisms of Marxian political economy tangible and interactive: specifically, the process by which technology investment displaces living labor, drives down the cost of producing a unit of output, and generates competitive market advantage — while simultaneously raising the organic composition of capital and suppressing the rate of profit.

The simulator operates at the level of two competing firms within a single commodity market, holding demand constant (inelastic). Its purpose is pedagogical and analytical: to demonstrate the structural logic of competitive accumulation and its consequences for workers, firms, and profitability. The cost accounting framework follows manufacturing absorption costing conventions, making the model applicable to anyone familiar with industrial production economics as well as to readers approaching it from political economy.

The tool draws on three interlocking bodies of analysis from Capital:

- Volume I, Chapter 10: The Working Day — the struggle over the length of the working day as a site of class conflict and the physiological limits of labor extraction.
- Volume I, Chapter 15: Machinery and Large-Scale Industry — how mechanization increases relative surplus value, displaces workers, and compels the extension of the working day.
- Volume III, Chapters 13–15: The Law of the Tendency of the Rate of Profit to Fall — how the rising organic composition of capital (C/v) suppresses the rate of profit even as the mass of surplus value grows.

2. Theoretical Framework

2.1 The Labor Theory of Value

The model is grounded in Marx's labor theory of value. Value is created only by living labor. Capital (machinery, technology, fixed assets) does not create new value; it transfers its own value to the product as it is consumed in production. This is the distinction between variable capital (v) and constant capital (C). Variable capital is the source of surplus value: workers produce more value than they are paid. Constant capital merely transfers pre-existing value without generating new value.

2.2 Output as a Function of All Production Inputs

A central design principle is that output is grounded in actual production inputs. Every widget produced is a direct product of how many workers are employed, how long they work each day, how many days per week they work, and the productivity multiplier conferred by technology:

```
output = workers × hours/day × days/week × 52 × tech multiplier
```

Each labor-hour produces exactly one widget at a tech multiplier of 1. The tech multiplier scales output per labor-hour proportionally. All four production inputs multiply together — no single input dominates. Market share falls out of production volume, not from assumed competitive dominance.

2.3 Absorption Costing and Fixed Overhead

The model uses manufacturing absorption costing. Annual fixed cost — depreciation (C / lifespan) plus variable capital (v) — is fully incurred regardless of how many units are sold or what the utilization rate is. This reflects the reality of capital-intensive production: depreciation is a fixed overhead charge that does not flex with output or sales volume.

This creates two distinct per-unit cost metrics that diverge whenever utilization falls below 100%:

```
Production cost / widget = total annual cost / output capacity Absorbed
cost / widget sold = total annual cost / units sold
```

Production cost per widget is the cost at full utilization — the exit threshold. If a firm cannot cover this at market price, it cannot survive even at full capacity. Absorbed cost per widget sold is the true per-unit burden when fixed overhead is spread only over units actually sold. When utilization is 60%, absorbed cost is 67% higher than production cost. A firm can sit above the exit threshold on production cost but still be unprofitable on an absorbed basis — the underutilization penalty.

This distinction is well understood in manufacturing cost accounting and is the mechanism by which capital-intensive firms are disproportionately exposed to demand downturns. A high-fixed-cost producer that loses market share does not simply earn less revenue — it absorbs the same fixed cost over fewer units, driving absorbed cost per unit up and compressing margins even before reaching the production-cost exit threshold.

2.4 Price-Taking Firms and Market Allocation

Both firms are price takers. The market price is set exogenously and applies equally to both firms. Neither firm has pricing power. Market share is determined proportionally by each firm's share of total combined output:

```
output share = firm output / (Firm A output + Firm B output) units sold
= min(firm output, market size × output share)
```

If total combined output is less than or equal to market size, each firm sells everything it produces. If combined output exceeds market size, each firm sells its proportional share of market demand. In this model, unsold units are simply not produced — there is no inventory stockpile. Underutilization means those units are never made, and the fixed annual cost is spread only over what was actually sold.

2.5 The Organic Composition of Capital and the Rate of Profit

As firms invest in labor-saving technology, C rises relative to v. Capital stock is derived from the variable capital base, the tech multiplier, and the capital lifespan:

```
C = tech multiplier × v × capital lifespan annual depreciation = C /
lifespan = tech multiplier × v organic composition (C/v) = tech
multiplier × capital lifespan
```

The rate of profit uses the Vol. III stock-based formulation — total capital advanced, not revenue or annual cost:

```
surplus value (s) = revenue − total annual cost rate of profit = s / (C
+ v)
```

As C rises relative to v, the denominator grows faster than the numerator. The individual firm gains a short-term competitive advantage by reducing cost per unit. But as the technology generalizes across the industry, all firms see their organic composition rise, and the industry-wide rate of profit falls. Individual rationality produces collective self-undermining of profitability.

2.6 The Working Day and Physiological Limits

Labor is expressed in physiological terms: hours per day and days per week. This makes the human cost of labor intensity immediately legible. Extending hours or days increases output volume but does not reduce production cost per widget, because cost and output scale

together. The physiological thresholds are flagged explicitly: amber above 10 hrs/day or 50 weekly hours; red above 12 hrs/day or 60 weekly hours.

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3. Model Structure and Formulas

3.1 Variable Capital (v)

```
v = workers × hours/day × days/week × 52
```

v is the total annual living labor index, expressed as hours rather than dollars. This keeps the focus on the quantity of labor time rather than nominal wage dynamics, consistent with Marx's value theory operating in units of socially necessary labor time.

3.2 Output

```
output = v × tech multiplier = workers × hours/day × days/week × 52 ×
tech multiplier
```

Each labor-hour produces one widget at multiplier = 1. All four production inputs multiply together. Output is the firm's production capacity — the maximum it can produce. Actual units sold may be less if total supply exceeds market size.

3.3 Capital Stock and Annual Cost

```
C (total capital stock) = tech multiplier × v × capital lifespan annual
depreciation = C / lifespan = tech multiplier × v total annual cost =
depreciation + v = (tech multiplier + 1) × v
```

Total annual cost is incurred in full every year regardless of utilization. It is the sum of the annual depreciation charge (fixed capital transferring value to the product) and variable capital (living labor). Capital stock is derived from the labor base, tech multiplier, and lifespan — preserving the internal logic that rising tech investment always implies rising organic composition.

3.4 Production Cost Per Widget

```
production cost / widget = total annual cost / output
= (tech multiplier + 1) / tech multiplier                          = 1 +
(1 / tech multiplier)
```

Production cost per widget is purely a function of the tech multiplier — independent of workers, hours, or days. This is the exit threshold: if market price falls below this, the firm cannot cover costs even at full utilization and must exit. This is also the floor the tech multiplier asymptotically approaches but never reaches: at multiplier = 1, cost/unit = 2.0; at multiplier = 10, cost/unit = 1.1; at multiplier = 20, cost/unit = 1.05.

3.5 Absorbed Cost Per Widget Sold

```
absorbed cost / widget sold = total annual cost / units sold
underutilization penalty = absorbed cost − production cost
```

Absorbed cost per widget sold is the true per-unit burden inclusive of all fixed overhead, spread only over units actually sold. When utilization is 100% it equals production cost. When utilization is below 100%, it is higher — the underutilization penalty. A firm can be above the exit threshold on production cost but report a negative absorbed margin if utilization is low enough. This is the manufacturing cost accounting reality the model makes explicit.

3.6 Market Allocation

```
total supply = Firm A output + Firm B output  (profitable firms only)
firm share = firm output / total supply units sold = min(firm output,
market size × firm share)
```

Unsold units are not produced — no inventory. Fixed annual cost is still fully incurred. The utilization rate (units sold / output capacity) drives the gap between production cost and absorbed cost.

3.7 Financial Metrics

```
revenue = market price × units sold profit = revenue − total annual
cost     (full fixed cost always incurred) profit margin = profit /
revenue rate of profit = s / (C + v)              (Vol. III stock-
based) breakeven month = total annual cost / (annual revenue / 12)
```

Profit is calculated against total annual cost — not cost of units sold. A firm that sells 60% of its output capacity earns revenue on 60% of output but pays 100% of its annual cost. Monthly figures are evenly distributed across 12 months for the chart series.

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4. Assumptions and Modeling Decisions

4.1 Core Assumptions

- Inelastic demand. Market size is fixed regardless of price.
- No wage rate. Labor expressed as hours index, not dollars.
- Single commodity market. Identical widgets; no product differentiation.
- Proportional market allocation. Market share determined by output share of total supply.
- No inventory. Unsold units are not produced. Fixed cost is still fully incurred.
- Absorption costing. Total annual fixed+variable cost is incurred regardless of utilization or sales.
- No financial structure. No interest, taxes, debt service, or dividends. Profit equals surplus value.
- No input cost variation. All cost variation flows through labor and capital.
- Straight-line depreciation. Capital stock depreciates evenly over its lifespan.
- No learning curves. Tech multiplier is fixed and does not improve with use or scale.
- Two-firm duopoly. No entry, exit equilibrium, or industry-wide convergence beyond two firms.
- Market exit is instantaneous. A firm whose production cost/unit exceeds market price produces nothing.
- Even monthly distribution. Annual revenue and cost are divided equally across 12 months for chart purposes.

4.2 Key Modeling Decisions

Absorption costing over variable costing

The model deliberately uses absorption costing rather than variable costing. In variable costing, fixed overhead would be treated as a period cost separate from per-unit cost. In absorption costing, it is fully absorbed into the unit cost. For a model focused on the competitive dynamics of capital-intensive manufacturing, absorption costing is the more realistic and analytically revealing choice: it surfaces the underutilization penalty, shows why high-fixed-cost firms are structurally exposed to market share loss, and connects directly to how manufacturers actually think about cost recovery.

Two distinct per-unit cost metrics

Separating production cost per widget (at full utilization) from absorbed cost per widget sold (at actual utilization) is the key innovation of this cost accounting structure. Production cost is the competitive and survival threshold — it is what determines whether a firm can exist in the

market at all. Absorbed cost is the operational reality — it is what the firm actually incurs per sold unit given how much of the market it captures. The gap between them is the underutilization penalty, and making it visible as a distinct metric is what allows the model to show how a firm can be technically solvent on production cost but still unprofitable in practice.

No inventory stockpile

The decision to treat unsold units as unproduced (rather than warehoused) simplifies the model and keeps the focus on the cost structure rather than inventory dynamics. In a full model, inventory would serve as a buffer between production and demand, allowing firms to overproduce in anticipation of future sales. Excluding this keeps the model clean and ensures that the absorbed cost calculation — spread over units sold, not units produced — directly reflects market absorption rather than production decisions.

Output grounded in all four production inputs

Output is the direct product of workers × hours/day × days/week × 52 × tech multiplier. Market share falls out of production volume. This is the only formulation consistent with the model's materialist premises: you cannot produce without labor and capital working together, and the quantity produced is determined by the quantity and intensity of both.

Rate of profit on capital stock, not revenue

Using s/(C+v) rather than profit/revenue as the rate of profit measure reflects Marx's Vol. III formulation. The accounting margin measures how efficiently a firm converts sales into profit. The Marxian rate measures the return on total capital advanced — which drives investment decisions at the macroeconomic level and tends to fall as capital deepens.

Monthly charts for one operating year

Charts are expressed as monthly cumulative figures across one operating year rather than as parameter sweeps. This reflects the model's primary analytical frame: a single production period in which fixed costs are committed and must be recovered. The breakeven month — when cumulative revenue crosses cumulative cost — is a standard manufacturing operations metric and is more immediately meaningful to practitioners than a parameter sweep.

5. User Inputs: Reference Guide

Each firm has five independent inputs. The market has two global inputs. All inputs have both a slider and a synchronized direct number entry field.

5.1 Market Inputs

Input	Range / Default	Description & Impact
Market price / widget	0.5–20	Exogenous price both firms face. Neither firm controls it. The revenue ceiling for both firms. Drag down to simulate competitive price destruction and find exit thresholds.
Market size (widgets/yr)	1k–10M / 500k	Total annual demand, fixed and inelastic. When combined firm output exceeds this, utilization falls below 100% and absorbed cost per unit rises above production cost.

Market price: effects of changing

- Increasing: raises revenue, surplus value, and rate of profit for profitable firms. Reduces competitive pressure.
- Decreasing: compresses margins. When market price crosses a firm's production cost/unit threshold, that firm exits. When it crosses both, crisis of valorization — neither firm can produce profitably.
- Note: a firm may show a negative absorbed margin before reaching the exit threshold. This means it is selling below its fully-absorbed cost per unit but above its production cost — it is technically solvent but operationally underwater.

Market size: effects of changing

- Increasing: raises revenue and profit for profitable firms proportionally. If combined output is already below market size, both firms sell everything they produce and utilization is 100%.
- Decreasing: if market size falls below combined output, utilization falls, absorbed cost per unit rises, and firms that looked profitable on production cost may become unprofitable on an absorbed basis.

5.2 Firm-Level Inputs

Input	Range / Default	Description & Impact
Workers	1–500 / A:100, B:40	Sets production scale. Does not affect production cost per widget. More workers = more output = potentially more market share.
Hours per day	1–24 / A:8, B:10	Length of the working day. Raises output and v proportionally. Physiological warnings: amber above 10 hrs/day, red above 12 hrs/day.
Days per week	1–7 / A:5, B:6	Working days per week. Raises output proportionally. Combined with hours/day determines weekly labor burden.
Tech multiplier	1–20 / A:2, B:7	The primary competitive variable. Lowers production cost/unit (= 1 + 1/mult). Raises output. Raises C and organic composition. The only input that simultaneously improves cost position.
Capital lifespan (yrs)	1–20 / A:5, B:10	Years the capital stock remains in service. Raises C = mult × v × lifespan without affecting annual cost or production cost/unit. Suppresses rate of profit by inflating the C+v denominator.

Workers

- Increasing: raises output, market share, revenue, and profit proportionally. Does not change production cost/unit or competitive position on cost.
- Decreasing: reduces scale across all metrics proportionally.
- Interaction with utilization: if combined output already exceeds market size, adding workers lowers utilization and raises absorbed cost per unit, even as absolute revenue may increase.
- For workers: more workers means more people subject to the working conditions set by other inputs. It also raises v, which proportionally raises C through C = mult × v × lifespan.

Hours per day

- Increasing: raises output, v, revenue, and profit proportionally. Production cost/unit unchanged. Does not improve competitive cost position.
- Decreasing: reduces all flow quantities proportionally.
- For workers: the most direct expression of the working day. Above 10 hrs: elevated injury risk and burnout. Above 12 hrs: chronic health damage, shortened life expectancy. The model flags but does not prevent these.
- Key insight: extending hours extracts more absolute surplus value but does not lower production cost per unit. Only the tech multiplier can do that. This is Marx's distinction between absolute and relative surplus value.

Days per week

- Increasing: same structure as hours per day — raises output and revenue, does not change cost/unit.
- Decreasing: reduces annual labor and all derived quantities proportionally.
- For workers: weekly hours = hours/day × days/week. Above 50: elevated strain. Above 60: severe overwork with documented chronic disease outcomes.

Tech multiplier

- Increasing: lowers production cost/unit (= 1 + 1/mult), raises output, raises C relative to v, and raises organic composition. The only input that simultaneously expands output AND reduces cost/unit.
- Decreasing: raises cost/unit. If cost/unit rises above market price, the firm exits.
- For workers: a higher multiplier means fewer workers are needed to produce the same output — the mechanism of technological unemployment and the reserve army of labor.
- Rate of profit paradox: raising the multiplier raises absolute surplus value but also raises C dramatically. The rate of profit s/(C+v) may fall even as absolute profit rises. The most technologically advanced firms tend toward the lowest rates of profit on total capital advanced.
- Absorbed cost interaction: a higher tech multiplier raises output, which — if the market can absorb it — raises utilization and brings absorbed cost closer to production cost.

Capital lifespan

- Increasing: raises C = mult × v × lifespan without changing annual depreciation or production cost/unit. Suppresses rate of profit by inflating the C+v denominator.
- Decreasing: reduces capital stock, raises rate of profit. Short lifespan implies rapid obsolescence — more characteristic of software or light manufacturing than heavy industry.
- For workers: no direct effect on working conditions. Indirectly, large long-lived capital stock creates pressure toward maximum utilization — longer shifts, more workers, continuous operation — to spread fixed cost over more units.
- Theoretical note: this is the mechanism by which industries with very long-lived capital — steel, mining, rail, utilities — tend structurally toward lower rates of profit than software or consumer goods.

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6. Output Metrics Reference

6.1 Labor

Output metric	What it means
Weekly hrs / worker	Hours/day × days/week. Amber above 50, red above 60.
Annual hrs / worker	Weekly hours × 52.
Widgets / worker / day	= hours/day × tech multiplier. Rises with tech multiplier only.
Widgets / day (firm)	Widgets/worker/day × workers. Total daily output capacity.
Widgets / week (firm)	Widgets/day × days/week. Total weekly output capacity.

6.2 Capital Composition

Output metric	What it means
v — variable capital	Annual living labor index (workers × annual hrs/worker).
C — total capital stock	mult × v × lifespan. Total capital advanced.
Annual depreciation	C / lifespan = mult × v. Annual fixed capital charge.
Annual cost (depr + v)	(mult + 1) × v. Total annual cost, always fully incurred.
Organic composition (C/v)	= mult × lifespan. Key structural indicator of capital intensity.
Labor share of annual cost	v / (depreciation + v). Falls as tech multiplier rises.

6.3 Production & Market

Output metric	What it means
Annual output (widgets)	Full production capacity: workers × annual hrs × tech multiplier.
Output share of supply	Firm output / total combined output. Determines market allocation.
Units sold	min(firm output, market size × output share). Actual sales.
Utilization rate	Units sold / annual output. Green ≥90%, amber ≥50%, red below 50%.
Unsold widgets	Output − units sold. Units that were not produced (no inventory in this model).
Production cost / widget	Total annual cost / output capacity. The exit threshold. = 1 + 1/mult.
Market price / widget	Exogenous market price.
Breakeven month	Month when cumulative revenue crosses cumulative cost. Green ≤ month 6, amber ≤ month 10, red if month 11–12.

6.4 Per Widget

Output metric	What it means
Market price / widget	Exogenous price. Same for both firms.
Production cost / widget	Total annual cost / output capacity. Exit threshold. Independent of utilization.
Absorbed cost / widget sold	Total annual cost / units sold. The true per-unit burden. Rises above production cost when utilization < 100%. The underutilization penalty made explicit.
Profit / widget (absorbed)	Market price − absorbed cost. The per-unit profit after full overhead absorption. Can be negative even when production cost < market price, if utilization is low enough.
Absorbed margin / widget	Profit/widget ÷ market price. The absorbed per-unit margin percentage.

6.5 Annual

Output metric	What it means
Revenue (index)	Market price × units sold.
Total annual cost (index)	Full annual cost, always incurred regardless of utilization or sales.
Surplus value (s)	Revenue − total annual cost.
Profit (index)	Equal to surplus value.
Profit margin	Profit / revenue. Green ≥15%, amber ≥0%, red if negative.
Rate of profit s/(C+v)	Marxian rate on total capital advanced. Tends to fall as C deepens.

6.6 Charts

Output metric	What it means
Cumulative revenue vs. cost	12 monthly data points. Solid line = revenue, dashed = cost, green = Firm A, blue = Firm B. Breakeven is where revenue crosses cost. Fixed cost line is straight — it does not flex with utilization.
Cumulative profit	Running profit total Jan–Dec. Zero line shown. Slope = monthly profit rate. A firm below zero at December is unprofitable for the year.
Annual snapshot	Revenue, total annual cost, and profit as bars for each firm. Profit bar turns red when negative.

7. Interpretive Notes and Scenarios

7.1 The Core Competitive Dynamic

Set Firm A with a low tech multiplier (2–3) and Firm B with a high multiplier (7–10). Firm B has lower production cost/unit and a larger output share. Drag the market price down. Watch Firm A's absorbed cost rise above market price before its production cost does — the utilization penalty hits first. Then watch production cost cross the threshold: Firm A exits.

7.2 The Underutilization Penalty

Set both firms with high tech multipliers and large workforces so combined output substantially exceeds market size. Watch utilization fall below 100% for both firms. Production cost/unit is unchanged, but absorbed cost/unit rises. Both firms may show positive production-cost margins while reporting negative absorbed margins and annual losses. This is the mechanism by which market saturation destroys profitability even in the absence of any change in technology or labor inputs.

7.3 Volume vs. Cost Competition

Give both firms the same tech multiplier but different worker counts. The larger firm captures more market share but at the same production cost/unit and the same absorbed margin — unless its added output pushes combined supply above market size, in which case utilization falls and absorbed cost rises for both. This demonstrates that volume competition and cost competition are structurally different: only the tech multiplier improves cost position.

7.4 Extending the Working Day

Give both firms the same tech multiplier. Increase Firm A's hours per day to 12 or 14. Production cost/unit does not change. Output rises, revenue rises, but absorbed cost per unit is unchanged at full utilization. If the added output pushes past market size and reduces utilization, absorbed cost actually rises. The physiological warnings activate. Workers are being destroyed and the firm gains no cost advantage.

7.5 The Rate of Profit Paradox

Give Firm B a tech multiplier of 15 and a capital lifespan of 15 years. Watch it dominate the market on cost and output share. Then look at its rate of profit — often single digits despite substantial absolute surplus. C is enormous relative to s. The firm wins competitively but offers a poor return on the capital advanced. This is the tendency of the rate of profit to fall: the mechanism that generates competitive advantage simultaneously undermines the return on the capital required to achieve it.

7.6 The Breakeven Month

Set Firm B with a high tech multiplier and large workforce such that it captures most of the market. Observe its breakeven month — the point where cumulative revenue crosses cumulative cost. Now reduce market size so Firm B's utilization falls. Watch the breakeven month shift later in the year as absorbed cost per unit rises. At low enough utilization, cumulative revenue never crosses cumulative cost — the firm is unprofitable for the full year despite a positive production-cost margin.

Source texts: Marx, K. Capital: A Critique of Political Economy, Volumes I–III. Harvey, D. The Limits to Capital (1982). Shaikh, A. Capitalism: Competition, Conflict, Crises (2016). Author: Steven Swamy — github.com/sswamy2000/marx-simulator